

PEAT REWETTING PLANNING SYSTEM

Entity Relationship Diagram

Entities, relationships and cardinalities of the PostGIS database, grouped by bounded context.

Project	Peat Rewetting Planning for 337 Peat Hydrological Units (KHG)
Document	Entity Relationship Diagram
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Introduction

This document shows the entities of the Peat Rewetting Planning database and how they relate. The diagrams are generated from the live database catalogue, so they match the implementation exactly. The snapshot covers 41 tables, 21 views and 55 foreign keys on PostgreSQL 17.5 with PostGIS 3.5.2.

How to read the diagrams

Notation	Meaning
<code>\ \ --o{</code>	One parent row has zero or many child rows; the child must have a parent
<code>o\ --o{</code>	Optional parent: the foreign key may be empty
PK, FK	Primary key, foreign key
Point, MultiPolygon, MultiLineString	PostGIS geometry type, always EPSG:4326 (longitude, latitude)
Entity with only <code>id</code>	Entity from another context, shown for context

To keep the diagrams readable, two groups of columns are left out:

- **Audit columns** present on every operational layer: `created_at` , `created_by` , `updated_at` , `updated_by` , `updated_via` , `rev` .
- **Lookup references** to code lists. They are listed in the section *Code lists*.

The full column list of every table is in the *Database Schema* document.

Overview

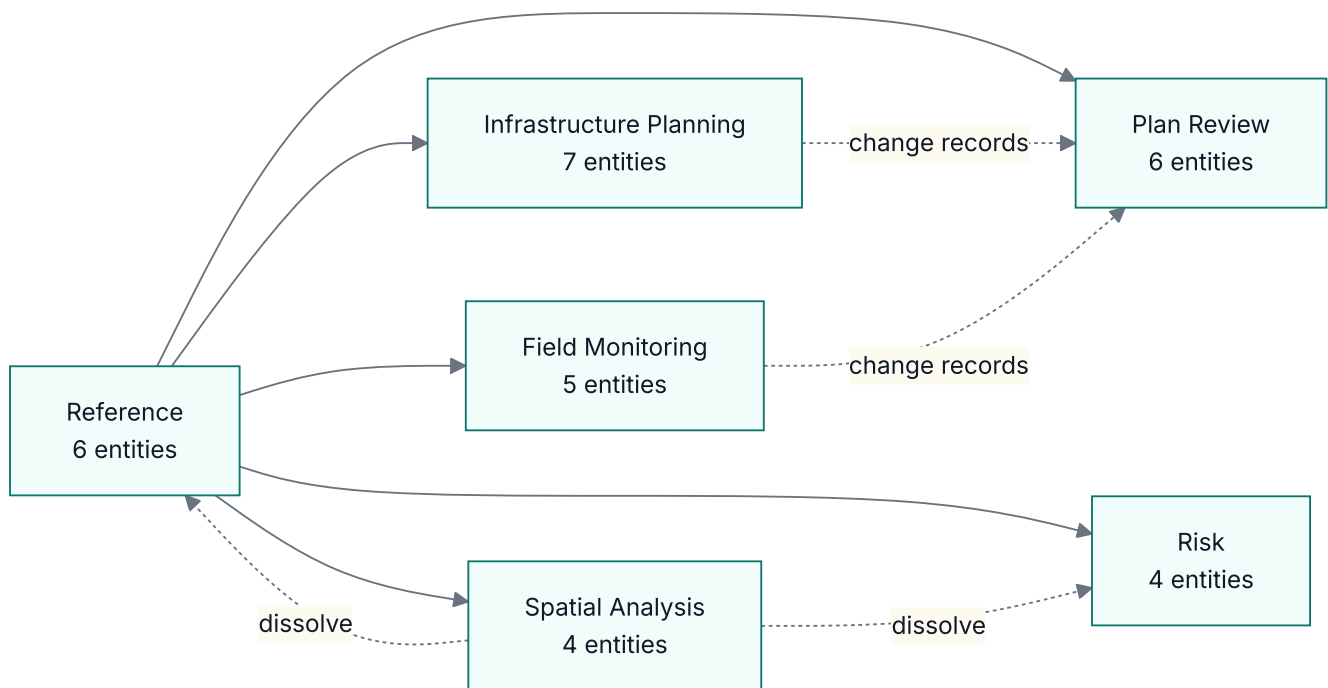


Figure 1. Contexts and their dependencies. Solid lines are foreign keys, dotted lines are data flows (dissolve during load, change records written by triggers).

Context	Entities	Rows
Reference	provinsi , kabupaten , kecamatan , desa , khg , perusahaan	9,065
Infrastructure Planning	kanal , sekat , sekat_konteks , sekat_bangunan , pompa , pintu_air , sungai	1,390,411
Risk	hotspot , areal_terbakar , ketebalan_gambut , konsesi	756,449
Field Monitoring	logger_tmat , tmat_bacaan , sumur_bor , sumur_debit , posko_karhutla	0
Spatial Analysis	unit_nasional , unit_target_2026 , kontur , kontur_lidar	1,964,745
Plan Review	khg_versi , perubahan , komentar , pengguna , template_poster , ekspor_poster	866

Reference

Administrative regions, KHG and companies. Shared by every other context.

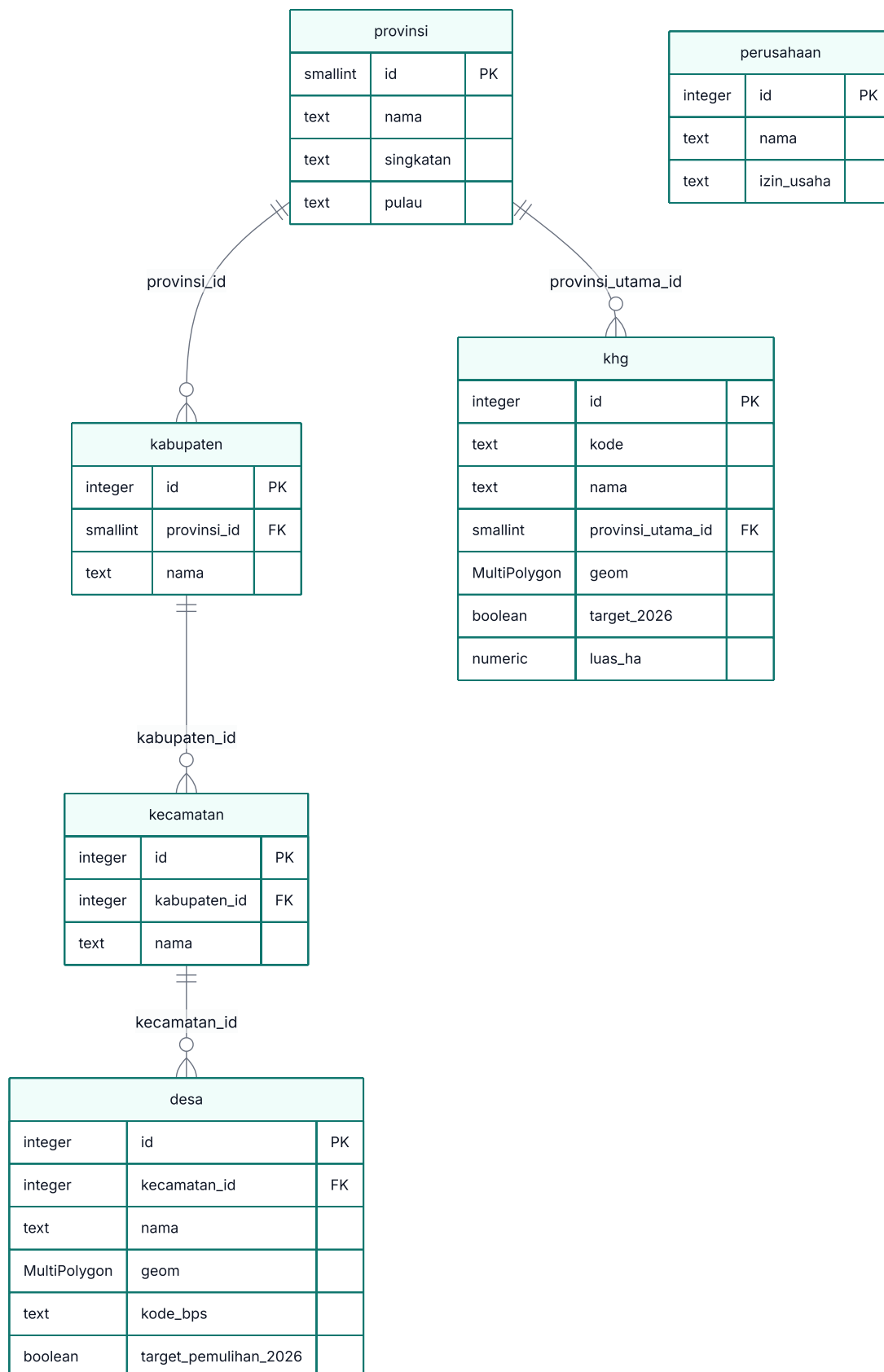


Figure 2. Reference entities.

Entity	Description	Rows
ref.provinsi	Province (38 rows, official Ministry of Home Affairs codes as primary key).	38
ref.kabupaten	Regency or city within a province.	142
ref.kecamatan	District within a regency.	892
ref.desa	Village. The 2,004 target villages for 2026 carry a boundary polygon.	5,860
ref.khg	Peat Hydrological Unit (KHG). Boundary dissolved from national analysis units.	866
ref.perusahaan	Company holding a plantation, forestry or mining permit.	1,267

Parent	Child	Foreign key	Cardinality
ref.provinsi	ref.kabupaten	provinsi_id	1 to many (required)
ref.kabupaten	ref.kecamatan	kabupaten_id	1 to many (required)
ref.kecamatan	ref.desa	kecamatan_id	1 to many (required)
ref.provinsi	ref.khg	provinsi_utama_id	1 to many (required)

Infrastructure Planning

Canals, canal blocks, pumps, gates and rivers that form a KHG plan.

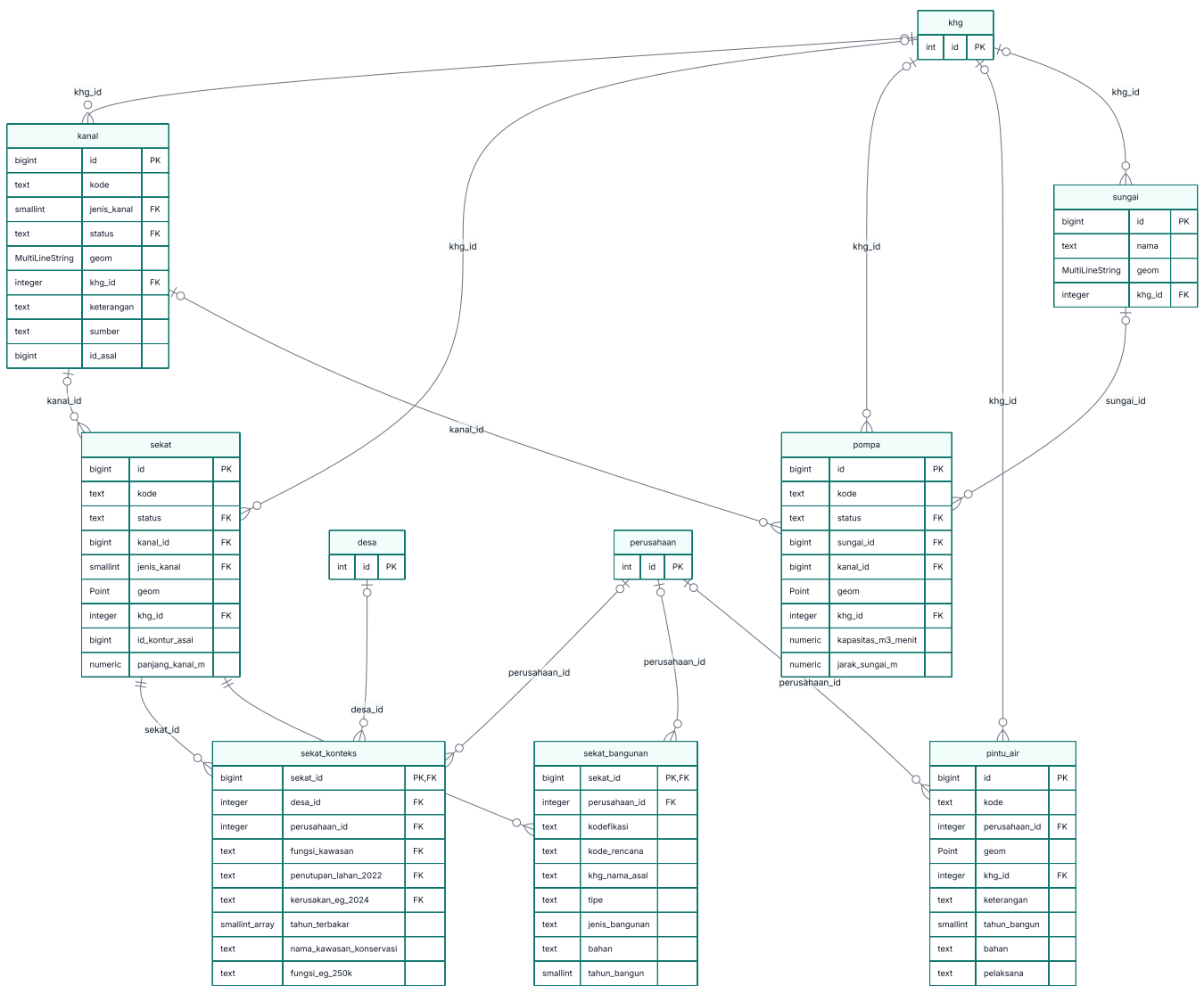


Figure 3. Infrastructure Planning entities.

Entity	Description	Rows
peta.kanal	Canal segment from the 2025 OpenStreetMap seamless canal layer. Editable.	657,621
peta.sekat	Canal block, planned or existing. Editable. Aggregate root.	366,371
peta.sekat_konteks	One to one overlay context of a planned canal block (village, permit, land cover, peat, burn history).	319,458

Entity	Description	Rows
peta.sekat_bangunan	One to one construction details of an existing canal block (budget, contractor, material, year).	46,913
peta.pompa	Planned or installed pump with suitability scores. Editable.	0
peta.pintu_air	Water gate. Editable.	48
peta.sungai	River line used as a water source for pumps. Editable.	0

Parent	Child	Foreign key	Cardinality
ref.khg	peta.kanal	khg_id	0..1 to many (optional)
peta.kanal	peta.sekat	kanal_id	0..1 to many (optional)
ref.khg	peta.sekat	khg_id	0..1 to many (optional)
ref.desa	peta.sekat_konteks	desa_id	0..1 to many (optional)
ref.perusahaan	peta.sekat_konteks	perusahaan_id	0..1 to many (optional)
peta.sekat	peta.sekat_konteks	sekat_id	1 to many (required)
ref.perusahaan	peta.sekat_bangunan	perusahaan_id	0..1 to many (optional)
peta.sekat	peta.sekat_bangunan	sekat_id	1 to many (required)
peta.kanal	peta.pompa	kanal_id	0..1 to many (optional)
ref.khg	peta.pompa	khg_id	0..1 to many (optional)
peta.sungai	peta.pompa	sungai_id	0..1 to many (optional)
ref.khg	peta.pintu_air	khg_id	0..1 to many (optional)
ref.perusahaan	peta.pintu_air	perusahaan_id	0..1 to many (optional)
ref.khg	peta.sungai	khg_id	0..1 to many (optional)

Fire hotspots, burned areas, peat thickness and concessions.

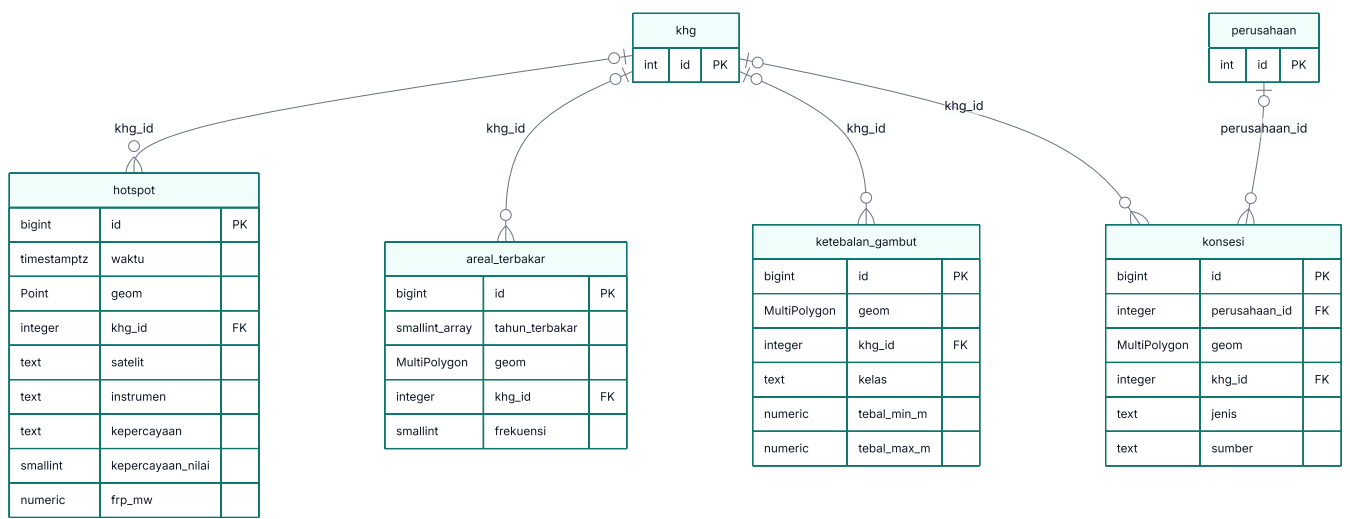


Figure 4. Risk entities.

Entity	Description	Rows
peta.hotspot	Satellite fire hotspot (MODIS or VIIRS), January to September 2026.	167,718
peta.area_terbakar	Burned area polygon with the list of burn years 2015 to 2026.	584,723
peta.ketebalan_gambut	Peat thickness class polygon per KHG.	2,938
peta.konsesi	Concession polygon per company, dissolved from analysis units.	1,070

Parent	Child	Foreign key	Cardinality
ref.khg	peta.hotspot	khg_id	0..1 to many (optional)
ref.khg	peta.area_terbakar	khg_id	0..1 to many (optional)
ref.khg	peta.ketebalan_gambut	khg_id	0..1 to many (optional)
ref.khg	peta.konsesi	khg_id	0..1 to many (optional)
ref.perusahaan	peta.konsesi	perusahaan_id	0..1 to many (optional)

Field Monitoring

Groundwater loggers, deep wells, fire posts and their time series.

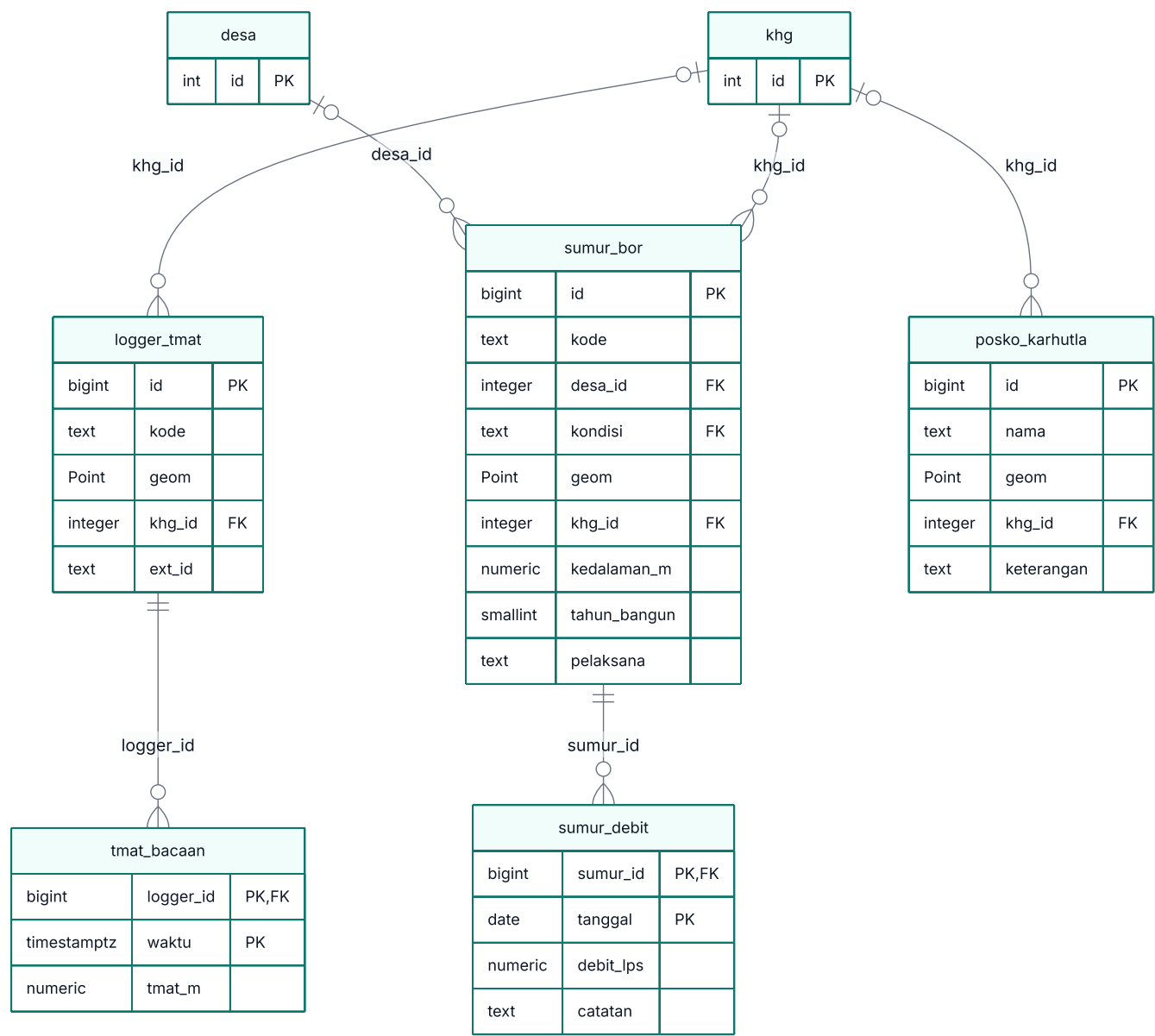


Figure 5. Field Monitoring entities.

Entity	Description	Rows
peta.logger_tmat	Groundwater level logger location. Editable.	0
peta.tmat_bacaan	Groundwater depth reading per logger and timestamp.	0
peta.sumur_bor	Deep well. Synchronised from the external Deep Well system. Editable.	0
peta.sumur_debit	Daily discharge of a deep well.	0

Entity	Description	Rows
peta.posko_karhutla	Forest and land fire post. Editable.	0

Parent	Child	Foreign key	Cardinality
ref.khg	peta.logger_tmat	khg_id	0..1 to many (optional)
peta.logger_tmat	peta.tmat_bacaan	logger_id	1 to many (required)
ref.desa	peta.sumur_bor	desa_id	0..1 to many (optional)
ref.khg	peta.sumur_bor	khg_id	0..1 to many (optional)
peta.sumur_bor	peta.sumur_debit	sumur_id	1 to many (required)
ref.khg	peta.posko_karhutla	khg_id	0..1 to many (optional)

Spatial Analysis

Read-only analysis units and contour lines.

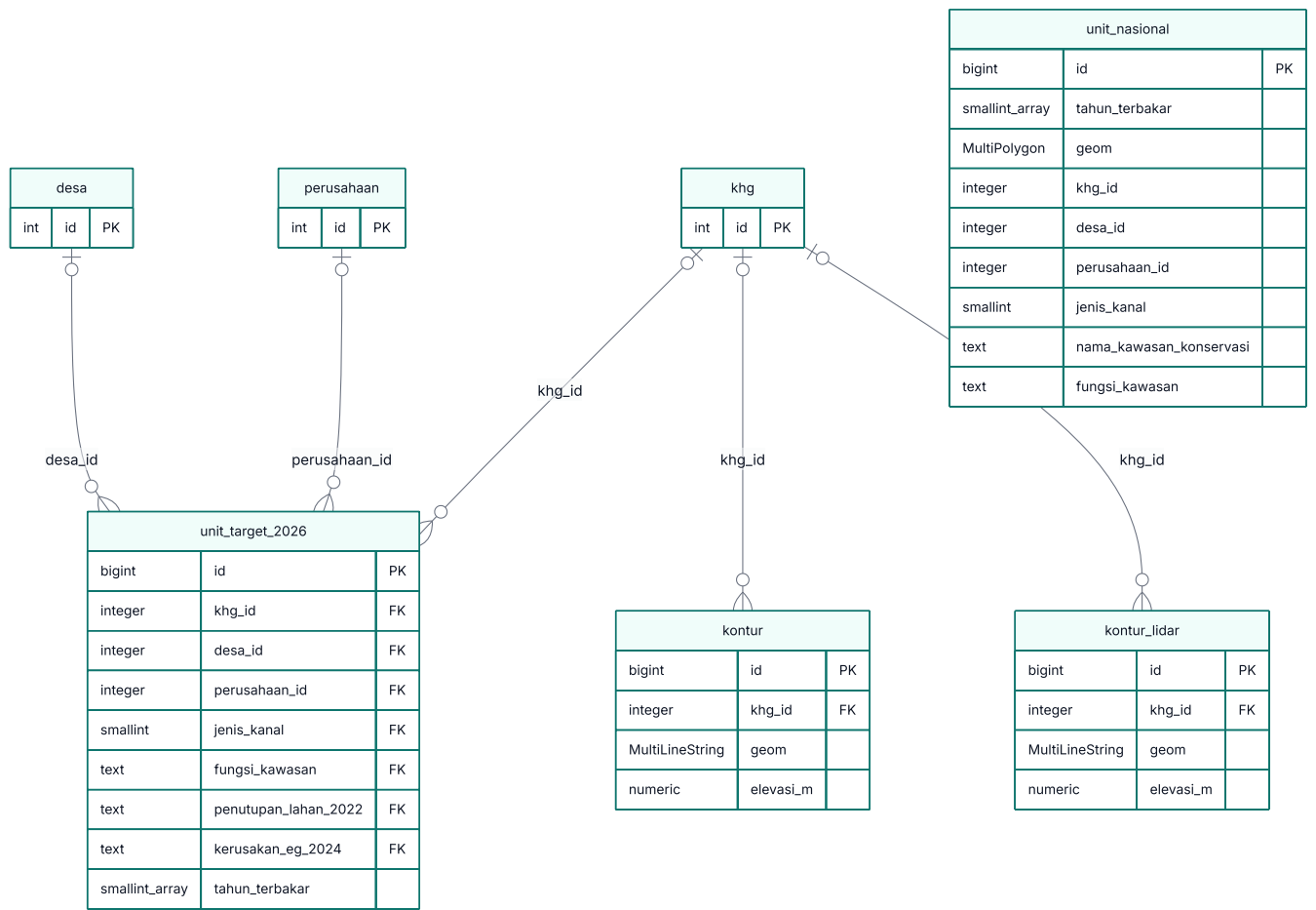


Figure 6. Spatial Analysis entities.

Entity	Description	Rows
analisis.unit_nasional	National analysis unit (overlay of KHG, village, permit, forest function, peat and burn history).	989,398
analisis.unit_target_2026	Analysis unit inside the 2,004 target villages for 2026.	789,336
analisis.kontur	Contour line from WorldDEM for 108 BRGM target KHG.	159,043
analisis.kontur_lidar	Contour line at 50 cm interval from LiDAR (Riau, Jambi, South Sumatra).	26,968

Parent	Child	Foreign key	Cardinality
ref.desa	analisis.unit_target_2026	desa_id	0..1 to many (optional)
ref.khg	analisis.unit_target_2026	khg_id	0..1 to many (optional)

Parent	Child	Foreign key	Cardinality
ref.perusahaan	analisis.unit_target_2026	perusahaan_id	0..1 to many (optional)
ref.khg	analisis.kontur	khg_id	0..1 to many (optional)
ref.khg	analisis.kontur_lidar	khg_id	0..1 to many (optional)

Plan Review

Plan versions, change records, comments and poster exports.

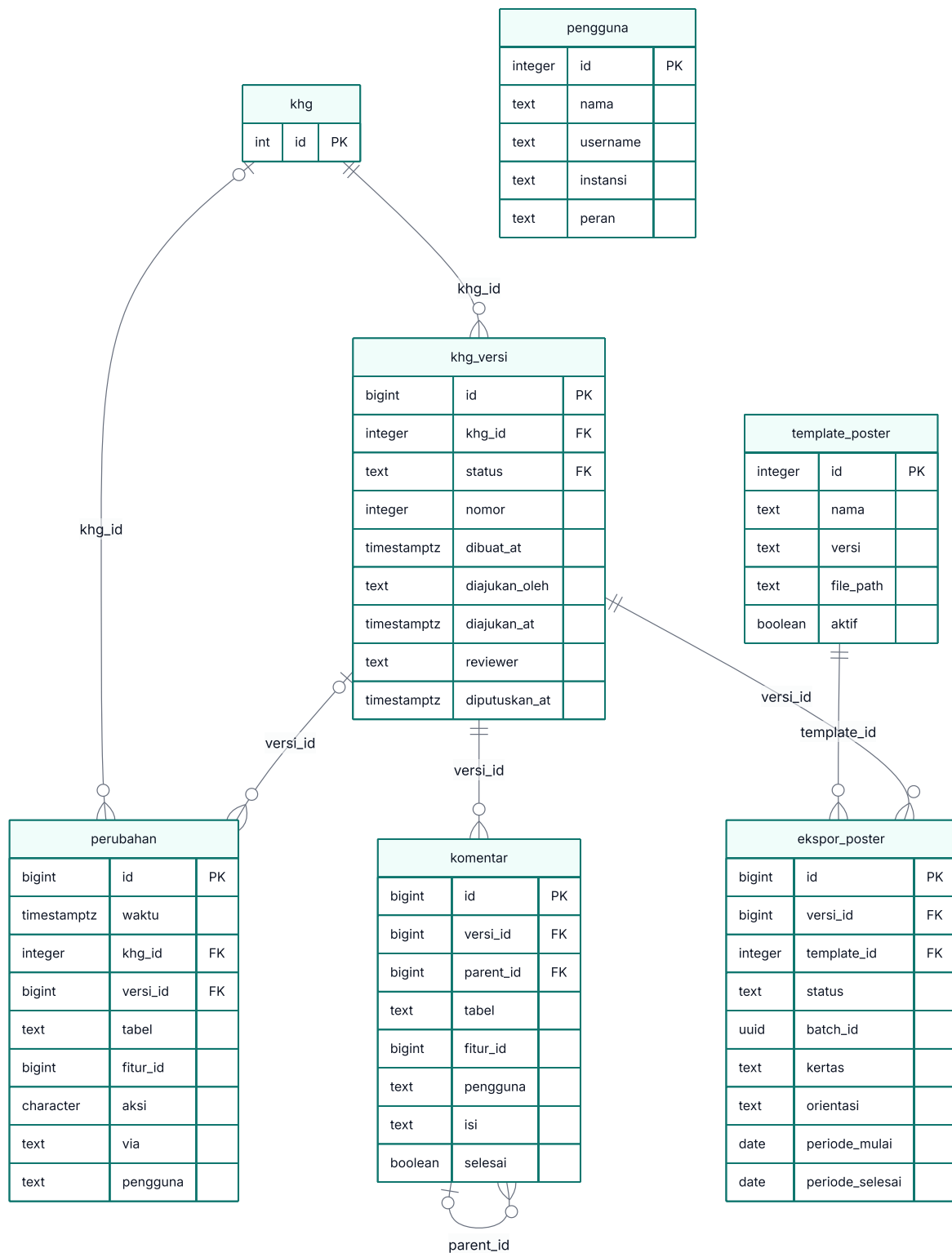


Figure 7. Plan Review entities.

Entity	Description	Rows
alur.khg_versi	Plan version of a KHG. Aggregate root of the review workflow.	866
alur.perubahan	Change record written by the audit trigger for every create, update and delete.	0
alur.komentar	Review comment on a version or on a single feature, with threaded replies.	0
alur.pengguna	Application user profile (name, agency, role).	0
alur.template_poster	Poster layout template (QGIS print layout file).	0
alur.ekspor_poster	Poster export job with paper size, progress and output file.	0

Parent	Child	Foreign key	Cardinality
ref.khg	alur.khg_versi	khg_id	1 to many (required)
ref.khg	alur.perubahan	khg_id	0..1 to many (optional)
alur.khg_versi	alur.perubahan	versi_id	0..1 to many (optional)
alur.komentar	alur.komentar	parent_id	0..1 to many (optional)
alur.khg_versi	alur.komentar	versi_id	1 to many (required)
alur.template_poster	alur.ekspor_poster	template_id	1 to many (required)
alur.khg_versi	alur.ekspor_poster	versi_id	1 to many (required)

Code lists

Code lists are small reference tables. They are used as foreign keys instead of PostgreSQL ENUM types, so QGIS can show them as drop-down lists and new values need only an INSERT.

Code list	Rows	Used by
ref.fungsi_kawasan	14	analisis.unit_target_2026.fungsi_kawasan , peta.sekat_konteks.fungsi_kawasan
ref.jenis_kanal	3	analisis.unit_target_2026.jenis_kanal , peta.kanal.jenis_kanal , peta.sekat.jenis_kanal
ref.kerusakan_eg	5	analisis.unit_target_2026.kerusakan_eg_2024 , peta.sekat_konteks.kerusakan_eg_2024
ref.kondisi_sumur	3	peta.sumur_bor.kondisi
ref.penutupan_lahan	26	analisis.unit_target_2026.penutupan_lahan_2022 , peta.sekat_konteks.penutupan_lahan_2022
ref.status_alur	5	alur.khg_versi.status
ref.status_kanal	3	peta.kanal.status
ref.status_pompa	3	peta.pompa.status
ref.status_sekat	4	peta.sekat.status

Public read model

The `api` schema contains views over the entities above. They join names from the reference context so that API users see readable values instead of ids. They have no relationships of their own.

View	Built from	Description
<code>api.khg_statistik</code>	hotspot, kanal, sekat, analysis units	Materialized view: expensive KHG statistics, refreshed daily.
<code>api.areal_terbakar</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.desa</code>	ref.desa and regions	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.hotspot</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.kanal</code>	peta.kanal and code lists	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.ketebalan_gambut</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.khg</code>	ref.khg, alur.v_khg_status, api.khg_statistik	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.konsesi</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.kontur</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.kontur_lidar</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.sekat</code>	peta.sekat, sekat_konteks, sekat_bangunan	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.sumur_bor</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.

View	Built from	Description
<code>api.unit_nasional</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.
<code>api.unit_target_2026</code>	peta or analisis table of the same name	Public view published as OGC API collection. Readable names instead of foreign keys.